

Dokumentasi Arsitektur Teknologi

E-Learning Bimbel Gambar Villa Merah

Versi	1.0
Tanggal	25 Juli 2026
Status	Production Ready

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Ringkasan Eksekutif

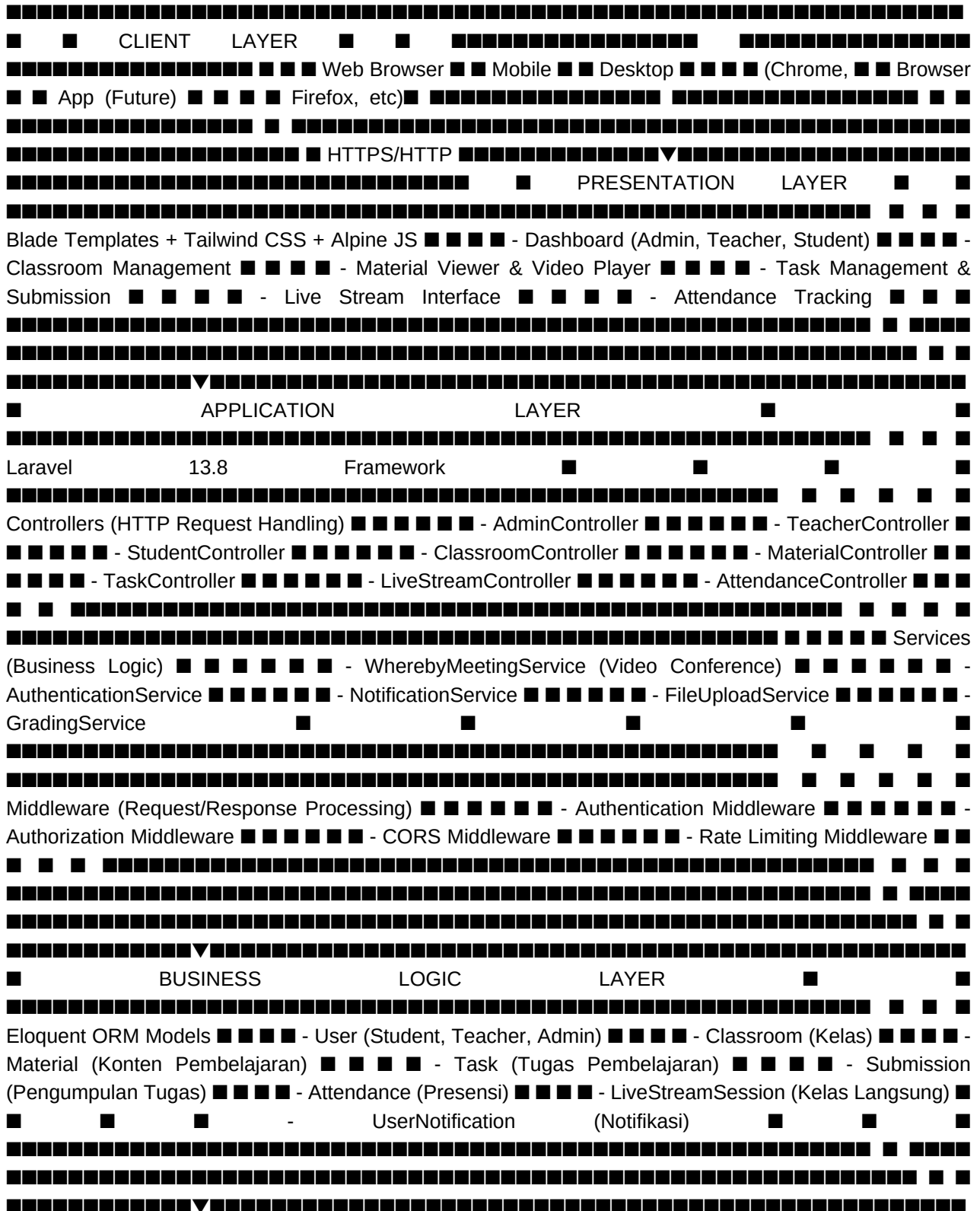
E-Learning Bimbel Gambar Villa Merah adalah aplikasi pembelajaran online (Learning Management System) yang dirancang khusus untuk mendukung pembelajaran menggambar dan pelajaran akademis (Skolastik) secara terintegrasi.

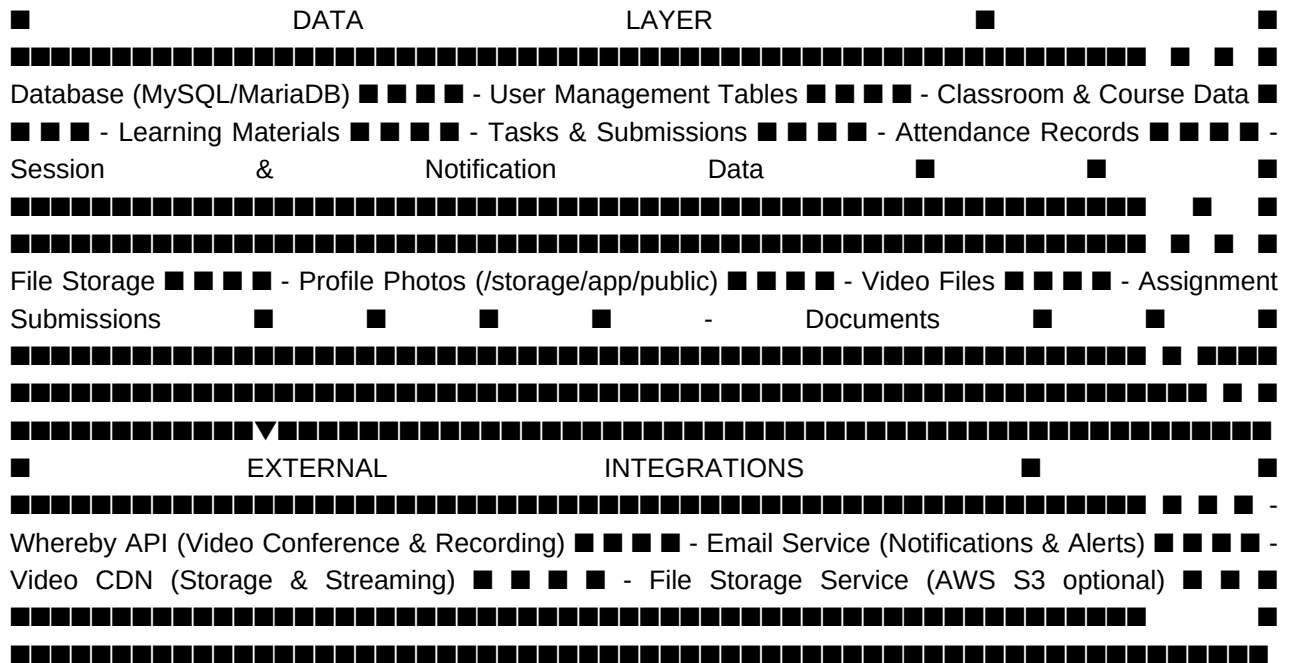
Karakteristik Utama:

- **Monolithic Architecture** berbasis Laravel 13.8
- **Multi-user System** dengan role-based access control (Admin, Teacher, Student)
- **Video Streaming** dengan adaptive bitrate
- **Live Session/Meeting** menggunakan Whereby API
- **Task Management** dengan submission dan grading
- **Attendance Tracking** real-time
- **Responsive UI** menggunakan Tailwind CSS

Arsitektur Sistem

Diagram Arsitektur Keseluruhan





Stack Teknologi

Backend

Frontend

Development

External Services

Struktur Aplikasi

Directory Structure

elearning-02/ ■■■■ app/ ■■■■ Http/ ■■■■ Controllers/ ■■■■ AdminController.php ■■■■ TeacherController.php ■■■■ StudentController.php ■■■■ StudentPageController.php ■■■■ StudentManagementController.php ■■■■ ClassroomController.php ■■■■ MaterialController.php ■■■■ TaskController.php ■■■■ AttendanceController.php ■■■■ LiveStreamController.php ■■■■ ProfileController.php ■■■■ Auth/ ■■■■ LoginController.php ■■■■ RegisterController.php ■■■■ AdminUserController.php ■■■■ Middleware/ ■■■■ Authenticate.php ■■■■ RedirectIfAuthenticated.php ■■■■ RoleMiddleware.php ■■■■ [Custom Middlewares] ■■■■ Models/ ■■■■ User.php ■■■■ Classroom.php ■■■■ Material.php ■■■■ Task.php ■■■■ Submission.php ■■■■ Attendance.php ■■■■ LiveStreamSession.php ■■■■ UserNotification.php ■■■■ Services/ ■■■■ WherebyMeetingService.php ■■■■ Providers/ ■■■■ AppServiceProvider.php ■■■■ routes/ ■■■■ web.php ■■■■ console.php ■■■■ [API routes] ■■■■ database/ ■■■■ migrations/ ■■■■ [Migration files] ■■■■ [Schema definitions] ■■■■ seeders/ ■■■■ factories/ ■■■■ UserFactory.php ■■■■ resources/ ■■■■ views/ ■■■■ layouts/ ■■■■ app.blade.php (Main layout) ■■■■ [Component layouts] ■■■■ auth/ ■■■■ login.blade.php ■■■■ register.blade.php ■■■■ dashboard/ ■■■■ admin.blade.php ■■■■ teacher.blade.php ■■■■ student.blade.php ■■■■ classroom/ ■■■■ materials/ ■■■■ tasks/ ■■■■ attendance/ ■■■■ live-stream/ ■■■■ welcome.blade.php ■■■■ css/ ■■■■ app.css (Tailwind styles) ■■■■ js/ ■■■■ app.js ■■■■ config/ ■■■■ app.php ■■■■ auth.php ■■■■ database.php ■■■■ filesystems.php ■■■■ logging.php ■■■■ mail.php ■■■■ session.php ■■■■ cache.php ■■■■ services.php ■■■■ queue.php ■■■■ storage/ ■■■■ app/ ■■■■ public/ (User uploads) ■■■■ framework/ ■■■■ views/ (Compiled blade views) ■■■■ logs/ (Application logs) ■■■■ public/ ■■■■ index.php (Entry point) ■■■■ build/ (Vite compiled assets) ■■■■ images/ (Logo & static images) ■■■■ storage -> ../storage/app/public ■■■■ bootstrap/ ■■■■ app.php ■■■■ providers.php ■■■■ tests/ ■■■■ Feature/ ■■■■ Unit/ ■■■■ TestCase.php ■■■■ vendor/ (Composer dependencies) ■■■■ node_modules/ (NPM dependencies) ■■■■ .env (Environment variables) ■■■■ composer.json ■■■■ package.json ■■■■ vite.config.js ■■■■ tailwind.config.js ■■■■ phpunit.xml ■■■■ artisan (Artisan CLI) ■■■■ README.md

Database Schema

Core Tables

Users Table

```
CREATE TABLE users ( id BIGINT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(255), email VARCHAR(255) UNIQUE, password VARCHAR(255), role ENUM('admin', 'teacher', 'student'), program_type ENUM('gambar', 'skolastik'), delivery_mode VARCHAR(255), video_accesses JSON, student_class VARCHAR(255), branch VARCHAR(255), academic_year VARCHAR(10), student_code VARCHAR(50), photo_path VARCHAR(255), email_verified_at TIMESTAMP NULL, approved_at TIMESTAMP NULL, approved_by BIGINT NULL, created_at TIMESTAMP, updated_at TIMESTAMP );
```

Classrooms Table

```
CREATE TABLE classrooms ( id BIGINT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(255), description TEXT, teacher_id BIGINT, academic_year VARCHAR(10), created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (teacher_id) REFERENCES users(id) );
```

Materials Table

```
CREATE TABLE materials ( id BIGINT PRIMARY KEY AUTO_INCREMENT, classroom_id BIGINT, title VARCHAR(255), description TEXT, content TEXT, video_url VARCHAR(255), file_path VARCHAR(255), order INT, is_published BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (classroom_id) REFERENCES classrooms(id) );
```

Tasks Table

```
CREATE TABLE tasks ( id BIGINT PRIMARY KEY AUTO_INCREMENT, classroom_id BIGINT, title VARCHAR(255), description TEXT, due_date DATETIME, max_score INT, created_by BIGINT, created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (classroom_id) REFERENCES classrooms(id), FOREIGN KEY (created_by) REFERENCES users(id) );
```

Submissions Table

```
CREATE TABLE submissions ( id BIGINT PRIMARY KEY AUTO_INCREMENT, task_id BIGINT, student_id BIGINT, file_path VARCHAR(255), submitted_at DATETIME, score INT NULL, feedback TEXT NULL, graded_by BIGINT NULL, graded_at DATETIME NULL, created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (task_id) REFERENCES tasks(id), FOREIGN KEY (student_id) REFERENCES users(id), FOREIGN KEY (graded_by) REFERENCES users(id) );
```

Attendance Table

```
CREATE TABLE attendance ( id BIGINT PRIMARY KEY AUTO_INCREMENT, classroom_id BIGINT, student_id BIGINT, session_date DATE, status ENUM('present', 'absent', 'late'), checked_at DATETIME, created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (classroom_id) REFERENCES
```

classrooms(id), FOREIGN KEY (student_id) REFERENCES users(id));

Live Stream Sessions Table

```
CREATE TABLE live_stream_sessions ( id BIGINT PRIMARY KEY AUTO_INCREMENT, classroom_id  
BIGINT, title VARCHAR(255), whereby_room_id VARCHAR(255), started_at DATETIME, ended_at  
DATETIME NULL, recording_url VARCHAR(255) NULL, status ENUM('scheduled', 'ongoing', 'completed'),  
created_at TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (classroom_id) REFERENCES  
classrooms(id) );
```

User Notifications Table

```
CREATE TABLE user_notifications ( id BIGINT PRIMARY KEY AUTO_INCREMENT, user_id BIGINT, type  
VARCHAR(255), title VARCHAR(255), message TEXT, data JSON, read_at DATETIME NULL, created_at  
TIMESTAMP, updated_at TIMESTAMP, FOREIGN KEY (user_id) REFERENCES users(id) );
```

Module dan Features

1. Authentication & Authorization

- **Login/Register** dengan email validation
- **Role-based Access Control** (Admin, Teacher, Student)
- **Session Management** menggunakan Laravel Session
- **Password Hashing** dengan bcrypt

File Utama:

- `app/Http/Controllers/Auth/LoginController.php`
- `app/Http/Controllers/Auth/RegisterController.php`
- `app/Http/Middleware/Authenticate.php`
- `routes/web.php`

2. User Management

- **Admin Dashboard** untuk manage users
- **User Approval System** untuk student registration
- **Profile Management** dengan photo upload
- **Role Assignment** (Admin, Teacher, Student)

File Utama:

- `app/Http/Controllers/AdminUserController.php`
- `app/Http/Controllers/ProfileController.php`
- `app/Models/User.php`

3. Classroom Management

- **Create Classroom** oleh teacher
- **Assign Students** ke classroom
- **Class Schedule** management
- **Student Enrollment** tracking

****File Utama:****

- `app/Http/Controllers/ClassroomController.php`
- `app/Models/Classroom.php`

4. Learning Materials

- ****Upload Materials**** (video, PDF, documents)
- ****Video Streaming**** dengan adaptive quality
- ****Material Organization**** per classroom
- ****Download Materials**** for offline learning

****File Utama:****

- `app/Http/Controllers/MaterialController.php`
- `app/Models/Material.php`

5. Task Management

- ****Create Tasks/Assignments**** oleh teacher
- ****Student Submission**** dengan file upload
- ****Grading System**** dengan scoring
- ****Feedback Provision**** untuk students

****File Utama:****

- `app/Http/Controllers/TaskController.php`
- `app/Models/Task.php`
- `app/Models/Submission.php`

6. Attendance Tracking

- ****Mark Attendance**** real-time
- ****Attendance Report**** per student
- ****Attendance History**** tracking
- ****Late/Absent**** recording

****File Utama:****

- `app/Http/Controllers/AttendanceController.php`
- `app/Models/Attendance.php`

7. Live Streaming / Video Conference

- ****Schedule Live Session**** untuk kelas
- ****Whereby Integration**** untuk meeting
- ****Session Recording**** automatic
- ****Participants Tracking**** real-time

****File Utama:****

- `app/Http/Controllers/LiveStreamController.php`
- `app/Services/WherebyMeetingService.php`
- `app/Models/LiveStreamSession.php`

8. Notifications

- ****Email Notifications**** untuk events
- ****In-app Notifications**** dashboard
- ****Assignment Reminders**** untuk students
- ****Grade Notifications**** when graded

****File Utama:****

- `app/Models/UserNotification.php`
- `config/mail.php`

9. Reporting & Analytics

- ****Student Performance Report****
- ****Class Statistics****
- ****Attendance Report****
- ****Assignment Submission Report****

Infrastruktur dan Deployment

Hosting Environment

[illegible]

Directory Structure on Server

```
/home/u930607946/ ■■■■ domains/ ■■■■ lmsvillamerah.sivmi.id/ ■■■■ public_html/ # Application root
■■■■ app/ ■■■■ bootstrap/ ■■■■ config/ ■■■■ database/ ■■■■ public/ ■■■■ resources/ ■■■■
■■■■ routes/ ■■■■ storage/ ■■■■ .env # Environment config ■■■■ composer.json ■■■■ [Laravel
files] ■■■■ .composer/ # Composer cache
```

Deployment Process

1. Version Control

Push perubahan ke GitHub

```
git add . git commit -m "Commit message" git push origin main
```

Pull di server

```
git pull origin main
```

2. Dependency Installation

Install PHP dependencies

```
composer install --no-dev --optimize-autoloader
```

Install Node dependencies

npm install npm run build

3. Laravel Setup

Clear cache

php artisan config:cache php artisan cache:clear php artisan route:cache php artisan view:cache

Run migrations

php artisan migrate

Seed database (if needed)

php artisan db:seed

4. File Permissions

chmod -R 755 storage chmod -R 755 bootstrap/cache chmod 644 .env

CI/CD Pipeline

GitHub Repository ↓ Webhook Trigger ↓ SSH to Server ↓ Git Pull ↓ Composer Install ↓ Cache Clear ↓
Application Ready

Keamanan

Authentication Security

- Password hashing dengan bcrypt
- Email verification untuk registration
- Session timeout configuration
- CSRF token protection untuk forms

Authorization Security

- Role-based access control (RBAC)
- Middleware protection per route
- Policy-based authorization
- Resource owner verification

Data Protection

- Encrypted database fields (password)
- HTTPS/SSL encryption in transit
- Prepared statements untuk prevent SQL injection
- Input validation dan sanitization

File Security

- File upload validation (size, type)
- Stored outside public directory
- Symlink untuk storage access
- Antivirus scanning for uploads

API Security

- Rate limiting per IP
- CORS configuration

- Request throttling
- API token validation

Monitoring

- Application logging
- Error tracking (Laravel Pail)
- Access logging
- Security audit trails

Performance dan Scalability

Optimization Strategies

1. Caching

// Config caching php artisan config:cache

// Route caching php artisan route:cache

// View caching php artisan view:cache

// Database query caching // Implemented in services

2. Database Optimization

- Indexed columns untuk frequent queries
- Lazy loading dengan pagination
- Query optimization
- Connection pooling

3. Frontend Optimization

- CSS minification via Vite
- JavaScript bundling & splitting
- Image optimization
- Lazy loading images

4. Codebase

- Dependency injection pattern
- Service layer pattern
- Repository pattern
- Model scoping

Scalability Considerations

Horizontal Scaling

- Stateless application design
- Session storage di database
- File storage di centralized location
- Load balancer ready

Vertical Scaling

- Database optimization
- Query caching
- View caching
- Asset optimization

Load Testing Guidelines

- Peak concurrent users: Expected 500+ users
- Database connection pooling
- CDN integration untuk static assets
- Video streaming dengan adaptive bitrate

Monitoring & Observability

Logging

Location: storage/logs/laravel.log Format: Single file with rotation Level: debug, info, warning, error, critical

Key Metrics

- HTTP request response time
- Database query performance
- Memory usage
- Disk space
- Error rate

Tools

- Laravel Pail untuk development logging
- Application error tracking

- User session monitoring
- API request logging

Technology Decision Matrix

Roadmap & Future Enhancements

Phase 1 (Current)

- ■ Core LMS features
- ■ User management
- ■ Classroom management
- ■ Material & task management
- ■ Live streaming integration

Phase 2 (Planned)

- [] Mobile app (React Native / Flutter)
- [] Advanced analytics & reporting
- [] AI-powered grading assistant
- [] Discussion forum
- [] Peer review system

Phase 3 (Future)

- [] Microservices architecture
- [] GraphQL API
- [] Real-time notifications (WebSocket)
- [] Advanced learning analytics
- [] Integration dengan external LMS

Support & Maintenance

Regular Maintenance Tasks

- Update Laravel & dependencies (quarterly)
- Database maintenance & optimization
- Security patches & updates
- Log rotation & cleanup
- Backup verification

Monitoring Checklist

- ☐ Application health check
- ☐ Database performance
- ☐ Disk space availability
- ☐ Error log monitoring
- ☐ User feedback tracking

Contact Information

- **Development:** jefrinickjr-prog (GitHub)
- **Repository:** <https://github.com/jefrinickjr-prog/lmsvillamerah>
- **Hosting:** Hostinger
- **Domain:** lmsvillamerah.sivmi.id

Appendix: Key Technologies

Laravel Framework

- **Version:** 13.8 (Latest LTS-compatible)
- **Components:** Routing, ORM, Auth, Mail, Cache
- **Documentation:** <https://laravel.com/docs>

Tailwind CSS

- **Version:** 3.4.19
- **Features:** Utility-first CSS, responsive design, dark mode
- **Documentation:** <https://tailwindcss.com>

Whereby API

- **Integration:** Video conferencing & recording
- **Authentication:** API key-based
- **Features:** Room management, recording, participants
- **Documentation:** <https://whereby.readme.io>

Vite Build Tool

- **Version:** 8.0.0
- **Features:** Hot module replacement, optimized builds
- **Performance:** Near-instant HMR
- **Documentation:** <https://vitejs.dev>

****END OF DOCUMENT****

Document Version: 1.0 *Last Updated: 25 Juli 2026* *Next Review: 25 Oktober 2026*